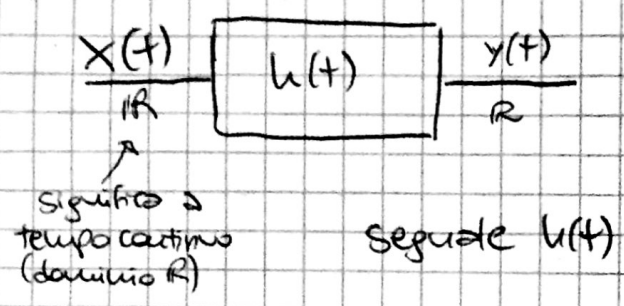


15/03/2018

FILTRO = trasformazione ~~da~~ di un segnale $x(t)$ di ingresso
che fornisce l'uscita $y(t)$



Relazione ingresso-uscita = integrale di convoluzione

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot h(t-\tau) \cdot d\tau = \int_{-\infty}^{+\infty} h(\tau) \cdot x(t-\tau) d\tau$$

$h(t)$ è la risposta del filtro

$$x(t) = \delta_R(t)$$

$$y(t) = \int_{-\infty}^{+\infty} h(\tau) \cdot \delta_R(t-\tau) d\tau = h(t)$$

se l'ingresso è un impulso
la risposta è il filtro
(RISPOSTA IMPULSIVA)

δ elemento neutro
della convoluzione

Il filtro è il modello di una trasformazione lineare (vale la sovrapposizione degli effetti)
e tempo invariante

$$x_1(t) \rightarrow y_1(t)$$

$$x_2(t) \rightarrow y_2(t)$$

$$a_1 \cdot x_1(t) + a_2 \cdot x_2(t) \rightarrow a_1 \cdot y_1(t) + a_2 \cdot y_2(t)$$

tempo invariante: non cambia le caratteristiche nel tempo, resta uguale l'uscita se c'è lo stesso ingresso

$$x(t) \rightarrow y(t)$$

$$x(t-t_0) \rightarrow y(t-t_0), \quad \forall t_0 \in \mathbb{R}$$

Segnale ritardato $\rightarrow$ uscita ritardata

1. è lineare (poiché la convoluzione è lineare)

2. è tempo invariante

DM: utilizzo la regola di traslazione della convoluzione

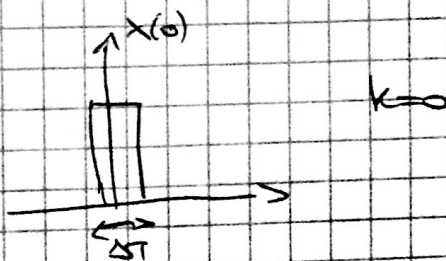
$$x(t-t_0) \rightarrow y(t-t_0)$$

NOTA:

$$x(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot \delta_R(t-\tau) d\tau$$

$$\approx \Delta T \cdot \sum_{k=-\infty}^{+\infty} x(k\Delta T) \cdot r_{\Delta T}(t-k\Delta T)$$

approssimo il segnale
con un andamento
costante a tratti



$r_{\Delta T}(t)$ chiamo $h_{\Delta T}(t)$
[fasi impulso = ^{grazi} risposta]

$$x(t) \approx \Delta T \cdot \sum_{k=-\infty}^{+\infty} x(k\Delta T) \cdot h_{\Delta T}(t-k\Delta T)$$

$$\approx \int_{-\infty}^{+\infty} x(\tau) \cdot h(t-\tau) \cdot d\tau$$

NOTAZIONE UNIFICATA

$$I = \mathbb{R}$$

$$\text{se } t \in \mathbb{R}, \quad -t \in \mathbb{R}$$

$$\text{se } t, t_0 \in \mathbb{R}, \quad t - t_0 \in \mathbb{R}$$

$$x(-t)$$

$$x(t - t_0)$$

$$\int_{-\infty}^{+\infty} x(t) dt \equiv \int_{\mathbb{R}} dt \cdot x(t)$$

$$\int_{\mathbb{R}} dt x(-t) = \int_{\mathbb{R}} dt x(t)$$

INVARIANTE
RISPETTO AL RIBALTAMENTO degli assi

$$\int_{\mathbb{R}} dt x(t - t_0) = \int_{\mathbb{R}} dt x(t)$$

INVARIANTE
RISPETTO ALLA TRASLAZIONE

$$\text{area } x(t) = \int_{\mathbb{R}} dt x(t)$$

$$\text{energia } x(t) = \int_{\mathbb{R}} dt |x(t)|^2$$

$$\text{Convulsione } x * y(t) = \int_{\mathbb{R}} d\tau x(\tau) \cdot y(t - \tau)$$

$$\text{impulso ideale } \delta_{\mathbb{R}}(t) = \delta(t)$$

$$\int_{\mathbb{R}} dt x(t) \cdot \delta_{\mathbb{R}}(t - t_0) = x(t_0) \quad \forall t_0 \in \mathbb{R}$$

$$\int_{\mathbb{R}} dt \delta_{\mathbb{R}}(t) = 1, \quad \int_{\mathbb{R}} \delta_{\mathbb{R}}(t) = \int_{\mathbb{R}} \delta_{\mathbb{R}}(-t)$$

I dominio temporale base:

$$\bullet \mathbb{R}$$

$$x(t), t \in I$$

$$\bullet \mathbb{Z}(T)$$

$$\text{se } t \in I, \quad -t \in I$$

$$\bullet \mathbb{R} / \mathbb{Z}(T_0)$$

$$t, t_0 \in I \Rightarrow t - t_0 \in I$$

$$\bullet \mathbb{Z}(T) / \mathbb{Z}(NT)$$

$$x(-t),$$

$$x(t - t_0), t \in I, t_0 \in I$$

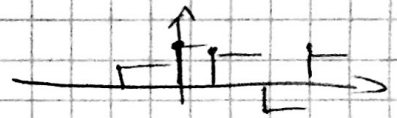
Sono seguiti ben
definiti per tutti
i domini

NOTAZIONE UNIFICATA:

$$\int_I dt \cdot x(t)$$

• se $I = \mathbb{R}$ $\int_{\mathbb{R}} dt \cdot x(t) = \int_{-\infty}^{+\infty} x(t) \cdot dt$ ($\geq$ T. CONTINUO)

• se $I = \mathbb{Z}(T)$ $\int_{\mathbb{Z}(T)} dt \cdot x(t) \stackrel{\text{def}}{=} T \cdot \sum_{k=-\infty}^{+\infty} x(kT)$ ($\geq$ T. DISCRETO)



• se $I = \frac{\mathbb{R}}{\mathbb{Z}(T_p)}$ $\int_{\frac{\mathbb{R}}{\mathbb{Z}(T_p)}} dt \cdot x(t) = \int_{t_0}^{t_0+T_p} x(t) dt$
(T. CONTINUO PERIODICO)

è un integrale in un periodo
qualsiasi
periodo = T_p

• se $I = \frac{\mathbb{Z}(T)}{\mathbb{Z}(NT)}$ $\int_{\frac{\mathbb{Z}(T)}{\mathbb{Z}(NT)}} dt \cdot x(t) = T \cdot \sum_{k=k_0}^{k_0+N-1} x(kT)$

(T. DISCRETO PERIODICO)

$$\int_I dt \cdot x(t) \quad \text{integrale di Haar}$$

area $x(t)$, $t \in I \stackrel{\text{def}}{=} \int_I dt \cdot x(t)$

energia $x(t)$, $t \in I \stackrel{\text{def}}{=} \int_I dt \cdot |x(t)|^2$

CONVOLUZIONE

$$z(t) = x * y(t), \quad t \in I \stackrel{\text{def}}{=} \int_I d\tau \cdot x(\tau) \cdot y(t-\tau)$$

es: $I = \mathbb{Z}(T)$, $t \equiv kT$, $\tau = h \cdot T$

$$z(kT) = T \cdot \sum_{h=-\infty}^{+\infty} x(hT) \cdot y(kT-hT)$$

NOTA: si verifica immediatamente che

$$\bullet \int_I dt \cdot x(t) = \int_I dt \cdot x(-t)$$

$$\bullet \forall t_0 \in I \quad \int_I dt \cdot x(t-t_0) = \int_I dt \cdot x(t)$$

Lineari:

$$\int_I dt \cdot (\alpha_1 \cdot x_1(t) + \alpha_2 \cdot x_2(t)) = \alpha_1 \int_I dt \cdot x_1(t) + \alpha_2 \int_I dt \cdot x_2(t)$$

$$x(t), y(t), t \in I$$

$$z(t) = x * y(t)$$

$$= \int_I d\tau \cdot x(\tau) \cdot y(t-\tau)$$

area convoluzione

$$\int_I dt \cdot \int_I d\tau \cdot x(\tau) \cdot y(t-\tau) =$$

$$= \int_I d\tau \cdot x(\tau) \cdot \left\{ \int_I dt \cdot y(t-\tau) \right\}$$

$$= \int_I d\tau \cdot x(\tau) \cdot \int_I dt \cdot y(t) = \text{area } x \cdot \text{area } y$$

impulso ideale $\delta_I(t)$

$$\bullet I = \mathbb{R}$$

T. CONTINUO

$$\delta_{\mathbb{R}}(t) = \delta(t)$$

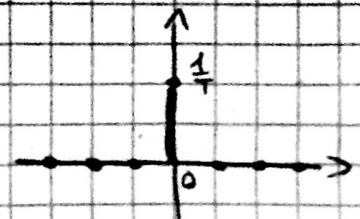
delta di Dirac



$$\bullet I = \mathbb{Z}(T)$$

$$\delta_{\mathbb{Z}(T)}(kT) = \begin{cases} \frac{1}{T} & \text{per } k=0 \\ 0 & \text{altrimenti} \end{cases}$$

T. DISCRETO

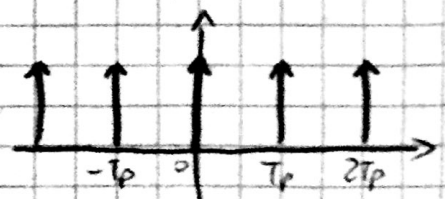


$$\bullet I = \frac{\mathbb{R}}{\mathbb{Z}(T_p)}$$

$$\delta_{\frac{\mathbb{R}}{\mathbb{Z}(T_p)}}(t) = \sum_{k=-\infty}^{+\infty} \delta_{\mathbb{R}}(t - kT_p)$$

T. CONTINUO PERIODICO

$$= \sum_{k=-\infty}^{+\infty} \delta(t - kT_p)$$



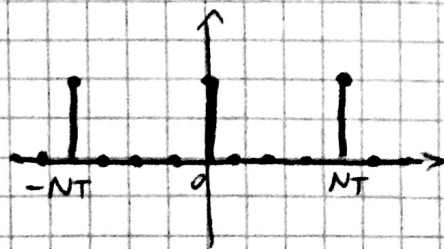
↳ TRENO di IMPULSI

$$\bullet I = \frac{z(t)}{z(NT)}$$

$$\delta_{\frac{z(t)}{z(NT)}}(kT) = \sum_n \delta_{z(t)}(kT - nNT)$$

periodizzazione

T. DISCRETO PERIODICO



si verifica immediatamente che:

$$\bullet \delta_I(t) = \delta_I(-t) \quad [e \text{ pari}]$$

$$\bullet \int_I dt \cdot s(t) \cdot \delta_I(t-t_0) = s(t_0)$$

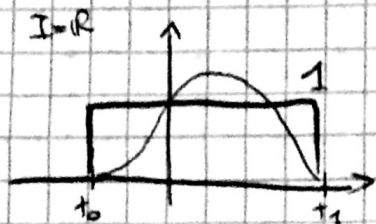
$\forall t_0 \in I$ $\rightarrow$ dominio base periodico

$$\bullet \text{area } \delta = \int_I dt \cdot \delta_I(t) = 1$$

$$\bullet x * \delta_I(t) = \delta_I * x(t) = x(t), \quad t \in I$$

[δ elemento neutro convoluzione]

Estensione e durata di un segnale



$$E_x = (t_0, t_1)$$

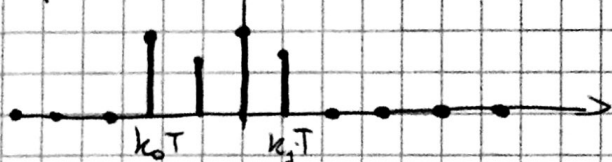
$$x(t) = 0 \text{ per } t \notin E_x$$

$$D_x = t_1 - t_0$$

$$I_{E_x}(t) = \begin{cases} 1 & \text{per } t \in E_x \\ 0 & \text{altrimenti} \end{cases}$$

$$D_x = \int_{-\infty}^{+\infty} I_{E_x}(t) dt$$

invece per $I = \delta(t)$

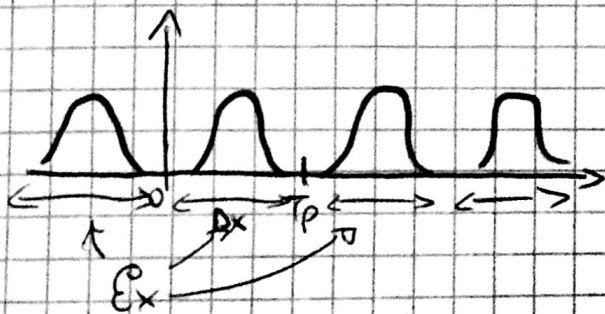


$$E_x = \{k_0 \cdot T, (k_0 + 1)T, \dots, k_N \cdot T\}$$

$$N \text{ costanti: } N = k_N - k_0 + 1$$

$$x(kT) = 0 \text{ per } kT \notin E_x$$

$$D_x = \int_{-\infty}^{+\infty} dt I_{E_x}(t) = T \sum_{k=-\infty}^{+\infty} I_{E_x}(kT) = T(k_N - k_0 + 1)$$



estensione e durata convoluzione

nel caso $I = z(t)$

$$z(t) = \int_{-\infty}^{+\infty} d\tau \cdot x(\tau) \cdot y(t-\tau), \quad t \in z(t)$$

$$t = kT, \quad \tau = lT$$

$$z(kT) = T \cdot \sum_{l=-\infty}^{+\infty} x(lT) \cdot y(kT - lT)$$

$$E_x = \{l_{0x} \cdot T, (l_{0x}+1)T, \dots, l_{1x} \cdot T\}$$

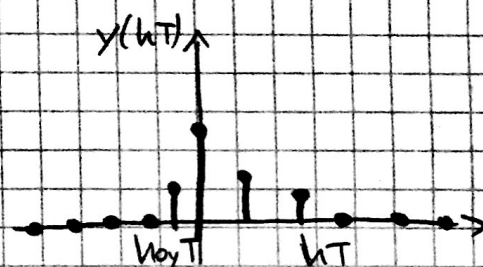
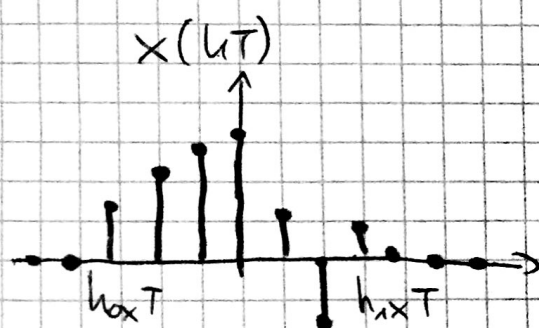
$$E_y = \{l_{0y} \cdot T, (l_{0y}+1)T, \dots, l_{1y} \cdot T\}$$

$$N = l_{1x} - l_{0x} + 1$$

$$D_x = NT$$

$$D_y = MT$$

$$M = (l_{1y} - l_{0y} + 1)$$



• se $kT - l_{0x}T < l_{0x}T$

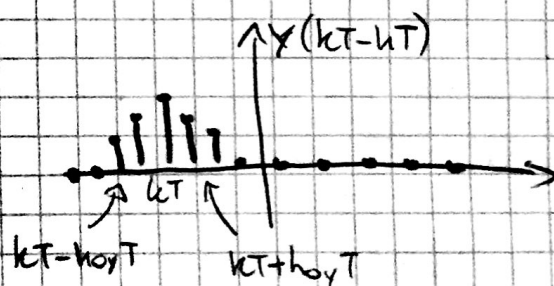
$$kT < l_{0x}T + l_{0y}T$$

$$z(kT) = 0$$

• se $kT - l_{1y}T > l_{1x}T$

$$kT > l_{1y}T + l_{1x}T$$

$$z(kT) = 0$$



$$E_x = \{(l_{0x} \cdot T + l_{0y} \cdot T), (l_{0x} + l_{0y} + 1)T, \dots, (l_{1x} + l_{1y}) \cdot T\}$$

$$\varepsilon_z = \varepsilon_x + \varepsilon_y$$

$$D_z = (h_{1x} + h_{1y} - h_{0x} - h_{0y} + 1) \cdot T$$

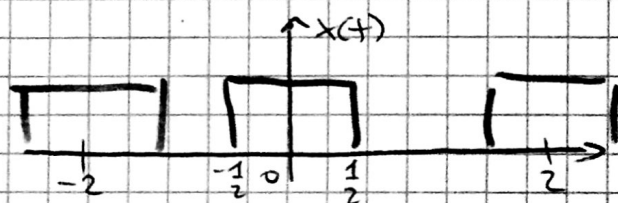
$$= D_x + D_y - T$$

in altre parole: • Se ε_x ha N costanti

ε_y ha M costanti

ε_x ha $M+N-1$ costanti

Esercizio: $I = R/z(z)$



$$x(t) = \sum_{k=-\infty}^{+\infty} \text{rect}(t - 2k)$$

$$y(t) = x(t)$$

$$z(t) = \int_{\frac{R}{z(z)}} d\tau \cdot x(\tau) \cdot y(t-\tau) = \int_{-1}^1 x(\tau) \cdot y(t-\tau) d\tau$$

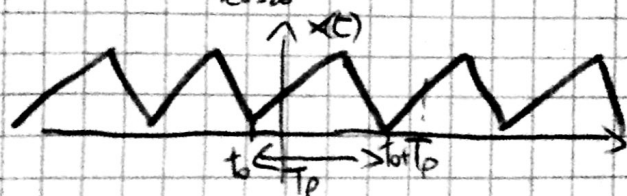
In generale

$$x(t) = \sum_{k=-\infty}^{+\infty} x_1(t - kT_p)$$

↑
ripetizione
periodica

$$x_1(t) = \begin{cases} x(t) & t \in (t_0, t_0 + T_p] \\ 0 & \text{altrove} \end{cases}$$

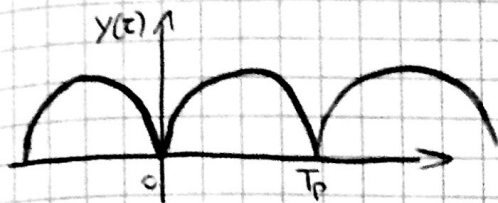
$$y(t) = \sum_{k=-\infty}^{+\infty} y_1(t - kT_p)$$



$$x(t) = \text{rip}_{T_p} \cdot x_a(t)$$

$$y(t) = \text{rip}_{T_p} \cdot y_a(t)$$

$$x * y(t) = \text{rip}_{T_p} (x_a * y_a(t))$$



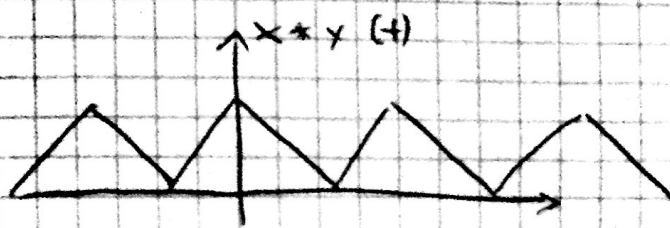
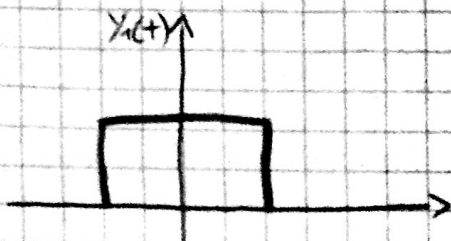
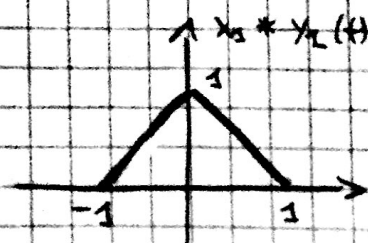
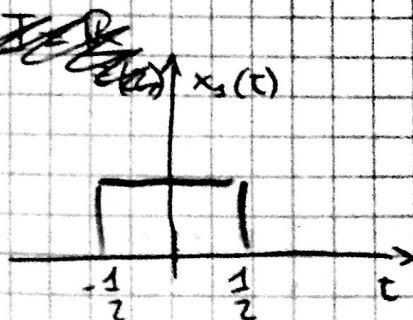
$$x(\tau) = \sum_{k=-\infty}^{+\infty} x_a(\tau - kT_p) = \text{rip}_{T_p} \cdot x_a(\tau)$$

$$y(\tau) = \sum_{k=-\infty}^{+\infty} y_a(\tau - kT_p) = \text{rip}_{T_p} \cdot y_a(\tau)$$

$$z(t) = \int_{\substack{\mathbb{R} \\ z(\tau)}} d\tau x(\tau) \cdot y(t-\tau)$$

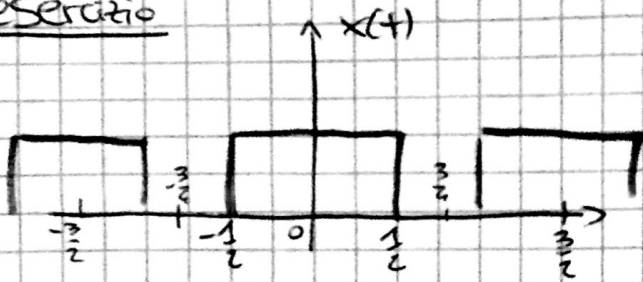
$$= \int_{t_0}^{t_0+T_p} x(\tau) y(t-\tau) d\tau = \int_{-a}^{+a} x_a(\tau) \cdot y(t-\tau) d\tau$$

$$= \int_{-a}^{+a} x(\tau) \cdot \left(\sum_{k=-\infty}^{+\infty} y_a(t-\tau-kT_p) \right) d\tau = \sum_{k=-\infty}^{+\infty} \int_{-a}^{+a} x_a(\tau) y_a(t-kT_p-\tau) d\tau$$



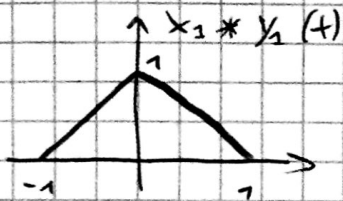
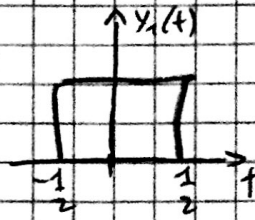
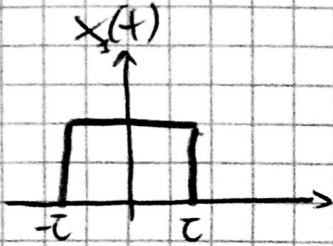
esercizio

$$t \in \frac{\mathbb{R}}{T\mathbb{Z}}$$



$$y(t) = x(t)$$

$$z(t) = x * y(t)$$



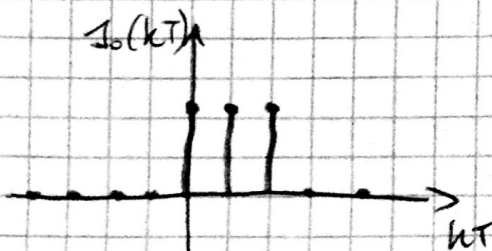
→ in blu la somma dei singoli triangolari

Esercizio

$$I = \mathbb{Z}(T)$$

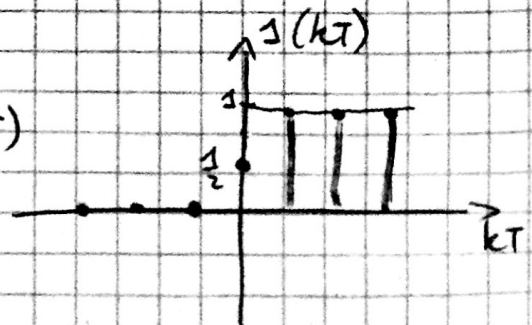
$$x(kT) = p^k \cdot 1_0(kT)$$

$$1_0(kT) = \begin{cases} 1 & \text{per } k > 0 \\ 0 & \text{altrimenti} \end{cases}$$

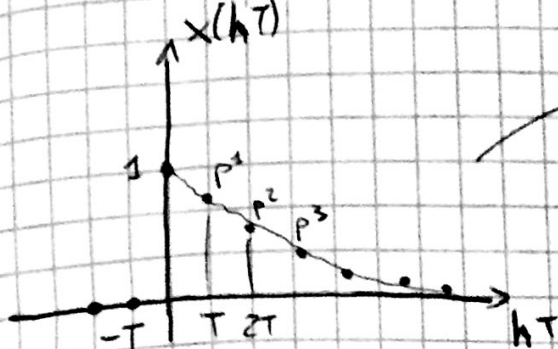


nel punto zero vale 1

$$1_0(kT) \neq 1(kT)$$



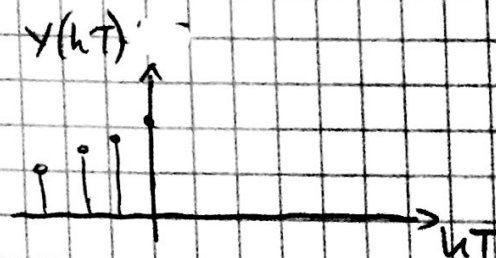
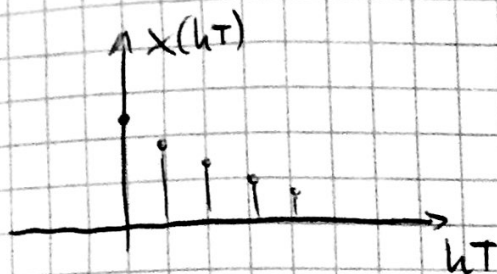
nel punto zero vale 1/2



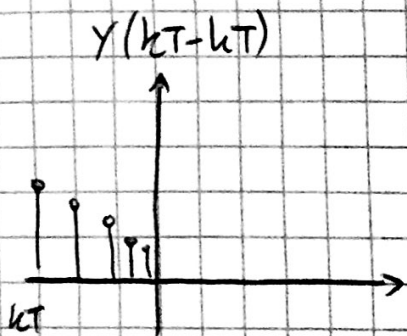
$p \in \mathbb{C}, |p| < 1$

$$x * y(kT)$$

$$y(kT) = x(-kT)$$



$$z(kT) = T \cdot \sum_{h=-\infty}^{+\infty} x(hT) \cdot y(kT - hT)$$



• per $k \leq 0$

$$z(kT) = T \cdot \sum_{h=0}^{+\infty} p^h \cdot p^{-(k-h)} = T \cdot p^{-k} \cdot \sum_{h=0}^{+\infty} p^{2h} = p^{-k} \cdot \frac{T}{1-p^2}$$

NOTA: Serie geometrica

$$\sum_{h=0}^{N-1} x^h = \frac{1-x^N}{1-x}$$

• per $k > 0$

$$z(kT) = T \cdot \sum_{h=k}^{+\infty} p^h \cdot p^{-(k-h)}$$

CAMBIO VARIABILE

$$h-k = h'$$

$$= T \cdot \sum_{h'=0}^{+\infty} p^{h'+k} \cdot p^{h'} = p^k \cdot \frac{T}{1-p^2}$$

$$\Rightarrow z(kT) = p^{|k|} \cdot \frac{T}{1-p^2}$$